APPROVED

FLIGHT MANUAL

FOR THE

AESL AIRTOURER T6 3M-2-54
3.570.

This Flight Manual is approved as the Manual referred to in the relevant Certificate of Airworthiness.

It is the responsibility of the pilot operating this aircraft to be familiar with the contents of this Flight Manual including all amendments thereto, and for ensuring that it is carried on all flights as required by Regulation 26 of the Civil Aviation Regulations 1953.

The Modification Appendix which forms part of this Flight Manual consists of operating information different from that contained in the remainder of the Flight Manual and is only applicable when the modifications as detailed are incorporated.

The pilot should therefore ascertain the applicability of the operating information contained in the Modification Appendix prior to flight in order to ensure the appropriate limitations can be complied with for the flight to be undertaken as required by Regulation 60 of the Civil Aviation Regulations 1953.

Amendments to this Flight Manual may only be incorporated by Aircraft Surveyors of the Civil Aviation Division of the Ministry of Transport unless delegation of this responsibility has been granted, in writing, to other persons.

If this Flight Manual is lost the Director of Civil Aviation should be informed at once.

Any person finding this Flight Manual should forward it immediately to the Director of Civil Aviation, Ministry of Transport, Wellington, New Zealand.

Date: 23 October 1970

for Director

AIR 72
Issue 2

CONTENTS

Compliance with the contents of this Flight Manual must be considered as being mandatory and overriding similar information contained in other documents such as the Owner's Manual or Pilot's Handbook.

However, as a considerable amount of valuable information is contained in such documents operators are advised to maintain them in a fully amended condition and use them, as appropriate, in conjunction with this Flight Manual.

SECTION 1 : GENERAL.

SECTION 2 : LIMITATIONS.

SECTION 3 : EMERGENCY PROCEDURES.

SECTION 4 : NORMAL PROCEDURES.

SECTION 5 : PERFORMANCE.

APPENDIX 1 : WEIGHT AND LOADING DATA.

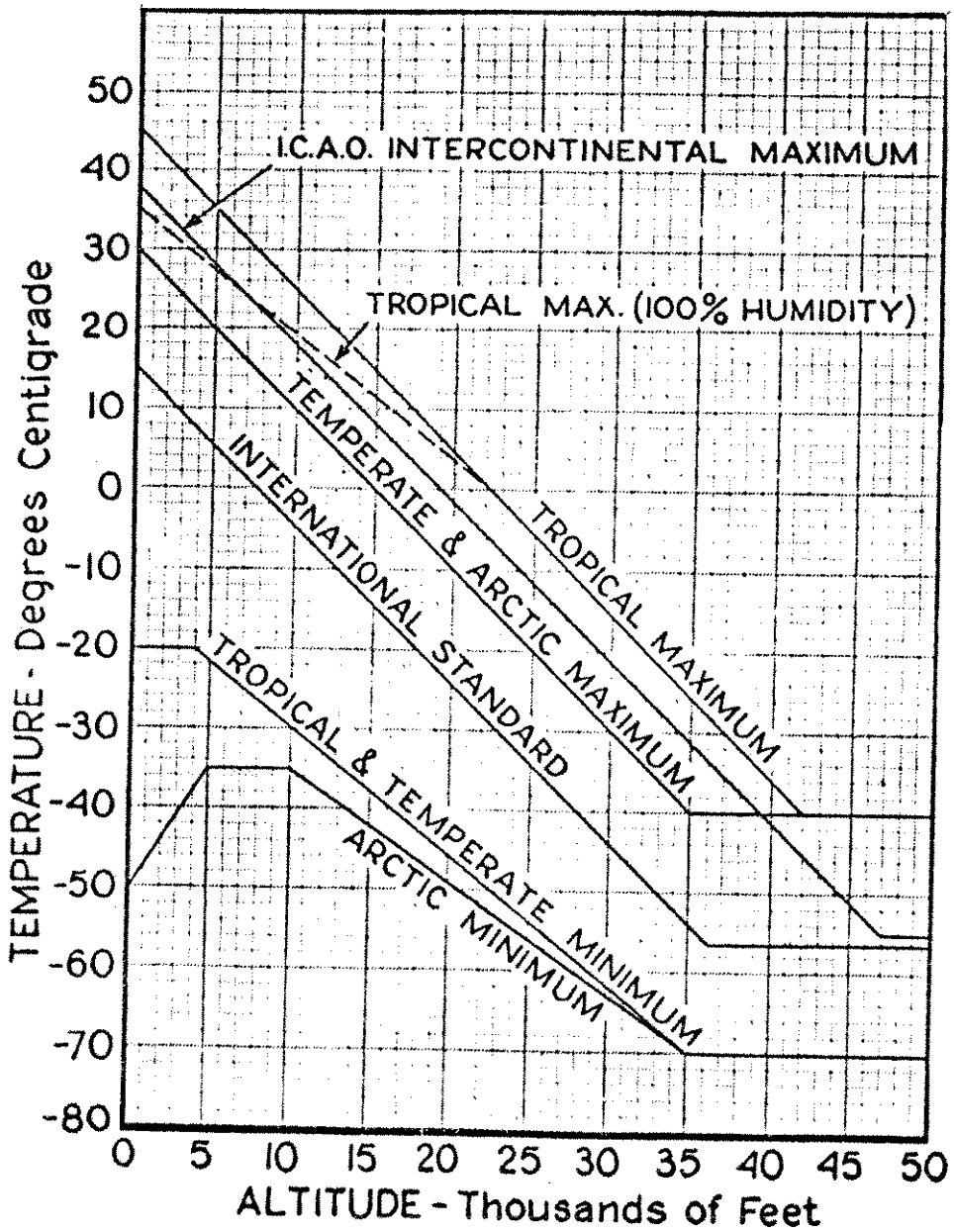
APPENDIX 2 : MODIFICATIONS.

SECTION 1 - GENERAL

CLIMATIC CONDITIONS

The operating suitability of the aeroplane has been established for temperatures up to the range defined by the ICAO Temperate and Arctic atmosphere (ISA + 15°C). This temperature range is shown in the Figure below.

No minimum temperature range has yet been established.



AMENDMENT RECORD SHEET

To record the incorporation of an amendment in this Flight Manual the person responsible should insert a brief description of the amendment pages affected, signature and date of incorporation in the appropriate columns adjacent to the amendment number.

Amendment Nr	Brief description of amendment	Pages Affected	Signature and date of incorporation of amendment
The initial issue of this Flight Manual consisted of Pages 1-27 inclusive.			
A/L 1	Forward C.G Limit Change - Spinning Restrictions. Pages 5, 7, 8, 16 + 24.	5, 7, 8 16 and 24	16-12-71.
A/L No 2.	To read: Mag drop should not exceed 175 rpm. As per Lycoming S.I. No 1132 A	Pg 13	10/9/77
AL NO 3.	Change Total Usable fuel from: 28.7 imp gals to 27.0 Imp gals. (S.B. 035 item 5)	Serial 2 Page 5	25/4/78.
A/L No. 4.	REPLACEMENT OF NARCO COM/NAV MARK 12B WITH KING KX-175B AND REWORK OF AUDIO DISTRIBUTION.	PAGES 26 & 27A	9/11/81.
A/L No. 5.	As per Amendment 4, The Narco COM/NAV mark 12B had been replaced with King KX-175B. ADF is included in the Aircraft Empty weight.	Pages 23, 25 and 27A	10/9/82

SECTION 2 - LIMITATIONS

This aeroplane must be operated as a Semi-aerobatic Category or Aerobatic Category aeroplane in compliance with the operating limitations and instructions stated in this Flight Manual or required to be stated and displayed in the form of placards or markings.

AIRFRAME LIMITATIONS

Maximum Weight	:	Semi-aerobatic Category	: 1900 lb
		Aerobatic Category	: 1800 lb
Centre of Gravity Range:		Forward Limit	: 79.2 in. aft of datum at 1400 lb or less 82.5 in. aft of datum at 1900 lb. (Linear variation between the points given).
		Aft Limit	: 86.1 in. aft of datum at all weights.
Datum	:	50.0 in. forward of front face of engine bulkhead.	
Flight Load Factors	:	At weights not exceeding 1900 lb Flaps Up : + 4.4 to - 2.2. Flaps Down : + 2.2 At weights not exceeding 1800 lb Flaps Up : + 6 to - 3. Flaps Down : + 3.	

POWER PLANT LIMITATIONS

Maximum RPM	:	Take-off and continuous	: 2700 RPM
		Diving (at any throttle setting)	: 2700 RPM
Cylinder Head Temperature	:	Maximum take-off and diving	: 260°C (500°F)
		Maximum continuous operation	: 232°C (450°F)
Oil System	:	Oil Temperature :	Maximum, all conditions: 245°F (118°C) Minimum for continuous operation: 140°F (60°C)
		Oil Pressure :	Normal operation 60-90 PSI Maximum 100 PSI (Starting or Warm up) Minimum 25 PSI (Idling)
Fuel System	:	Use 80/87 minimum grade aviation fuel Total capacity : 29.0 Imp. gals Total Usable : ~~28.7 Imp.~~ gals 27.0 Imp gals. (AC3) S.B.035 item 3.	

INSTRUMENT MARKINGS

Instrument	Colour Code		Range or Limit
Oil Temperature	Yellow/Amber Arc Green Arc Red Line	(Caution Range) (Normal Operating Range) (Maximum)	5° - 50°C 50° - 118°C 118°C
Oil Pressure	Yellow/Amber Arc Green Arc Red Line	(Caution Range) (Normal Operating Range) (Maximum)	25 - 60 PSI 60 - 85 PSI 100 PSI
Tachometer	Green Arc Red Line	(Normal Operating Range) (Maximum)	500 - 2700 RPM 2700 RPM
Fuel Pressure	Yellow/Amber Arc Green Arc Red Line	(Caution Range) (Normal Operating Range) (Maximum)	0.5 - 2.0 PSI 2.0 - 5.0 PSI 5.0 PSI

AIRSPEED INSTRUMENT MARKINGS SEMI-AEROBATIC CATEGORY 1900 LB

Marking	Significance	Range or Limit
Red Radial Line	Never Exceed (Glide or dive, smooth air)	178 Knots
Yellow Arc	Caution Range	141 - 178 Knots
Green Arc	Normal Operating Range	55 - 141 Knots
White Arc	Flap Operating Range	49 - 103 Knots

AIRSPEED LIMITATIONS (CAS)

	Semi-Aerobatic Knots	Aerobatic Knots
Never Exceed Speed Vne	178	203
Maximum Structural Cruising Speed Vc	141	161
* Maximum Manoeuvring Speed Va	123	140
Maximum Speed Flaps Extended Vf	103	103
* NOTE: For manoeuvres involving an approach to stall conditions, or full application of rudder or aileron control.		

MISCELLANEOUS LIMITATIONS

Types of Operation

VFR, IFR, Day and Night Operations are permitted when the aircraft is equipped and the operations are carried out in accordance with the requirements of Civil Aviation Regulations 1953 and any other requirements issued pursuant thereto.

Flight in Icing Conditions

The aircraft is not approved for flight in airframe or propeller icing conditions.

Inverted Flying

The aircraft is not cleared for continuous inverted flying due to engine limitations.

Spinning

This is permitted at all weights up to 1900 lb but restricted to 1-1/2 turns to avoid exceeding airspeed limitations. No luggage may be carried.
Intentional spins with flaps extended are prohibited.

PLACARDS

The following placards must be prominently displayed:

(1) In front and in clear view of the pilot:

(a) "This aircraft must be operated as a semi-aerobatic or aerobatic category aircraft in compliance with the operating limitations stated in the form of placards, markings and manuals."

(b) "Airspeed indicator markings are for semi-aerobatic category. Refer to flight manual or placard for aerobatic category limitations."

(c) "This aircraft is approved for VFR day and night operations only." (If the required equipment is installed, IFR operations may be added to this placard.)

(d) "This aircraft is not approved for flight in airframe or propeller icing conditions."

(2) Close to the airspeed indicator:

Airspeed Limitations Knots IAS	Semi-aerobatic 1900 lb	Aerobatic 1800 lb
Never Exceed Vne	178	203
Maximum Structural Cruise Vc	141	161
Manouevring Speed Va	123	140
Maximum Flaps Extended Vf	103	103
Maximum Crosswind Landing or Take-off	20	20

(3) In clear view of the pilot:

(a) "~~Smoking is permitted except during refuelling, take-off, landing and aerobatics.~~" "No smoking".

(b) "Solo from LH seat only." (May be omitted if electrically operated flaps are installed.)

(c) "Spinning permitted at all weights up to one and a half turns only. Use normal recovery techniques. Intentional spins with flaps extended are prohibited."

(d) "The following manoeuvres are approved for semi-aerobatic and aerobatic categories. Speeds are recommended entry speeds Knots IAS.

Barrel roll 115; Slow roll 90; Aileron roll 100; Flick roll 80; Half roll and dive out 85; Half loop and roll out 140; Loop 130; Stall turn 130; Steep turn 1800 lb and below 140; Steep turn over 1800 lb 120; Inverted flight 100; Chandelle and lazy eight 110."

(4) Near the carburetter heat control:

"Take-off and climb: Full cold recommended.
Other conditions : Heated air as required particularly during approach and landing.

(5) In the baggage compartment:

"Baggage 100 lb maximum. Baggage must not be carried while performing aerobatics."

SECTION 3 - EMERGENCY PROCEDURES

ENGINE FAILURE AFTER TAKE-OFF

(1) Maintain at least 65 Knots (IAS).
(2) Aim to land straight ahead, turning only as necessary to avoid obstacles.
(3) Select best clear area for landing.
(4) If time permits, conduct a trouble check ascertaining switches ON, mixture FULLY RICH, fuel ON.
(5) Before landing ensure: IGNITION SWITCH - OFF.
FUEL - OFF.
(6) Avoid high obstacles.
Use flaps to best advantage.

FORCED LANDING

Should an engine fail other than immediately after take-off proceed as follows:

(1) Close the throttle, trim aircraft to speed for minimum gliding angle (85 Knots) (IAS).
(2) Select proposed landing area and proceed with forced landing plan.
(3) Conduct trouble check, ascertaining:

Fuel	-	ON
Fuel Pressure	-	At least 2 psi. If below switch on auxiliary fuel pump.
Carburetter Heat	-	HOT
Mixture	-	RICH
Ignition Switch	-	BOTH

(4) If the cause of the failure cannot be rectified or the failure is of an obvious permanent nature, continue approach and complete the following safety checks:

Hatches	-	CLOSED
Harness	-	TIGHT
Throttle	-	CLOSED
Fuel	-	OFF
Fire Extinguishers	-	READY FOR USE
Ignition Switch	-	OFF
Brakes	-	OFF

(5) When on the base leg employ the glide approach technique. If an excessive overshooting tendency is apparent, correction may be assisted by lowering full flap or executing a slipping turn on to the final approach.
(6) When certain that the selected landing area can be reached lower the flaps fully.

NOTE 1: Sideslipping with flaps down is permitted at speeds up to the maximum flaps extended speed.

NOTE 2: In simulating forced landings the engine should be cleared at least every 1,000 feet during the descent.

ACTION IN THE EVENT OF FIRE

IN FLIGHT

(1) Immediately fire is detected ensure:

Throttle	- CLOSED
Fuel	- OFF
	Raise nose to reduce air speed and RPM
Mixture	- IDLE CUT OUT
Switches	- Generator field switch OFF.

(2) Transmit emergency call before selecting master switch OFF.

(3) Select suitable landing area and make a forced landing as soon as possible.

(4) If the fire persists try the effect of sideslipping which, if effective, should be employed throughout the descent.

(5) Use the hand fire extinguisher to the best advantage, if necessary.

ON THE GROUND

(1) The following sequence of action is recommended:

Throttle	- CLOSED
Mixture	- IDLE CUT OUT
Ignition Switch	- OFF
Fuel	- OFF

(2) If necessary, release the brakes to enable the aircraft to be pushed away from burning fuel on the ground.

NOTE 1: Sealing off the carburetter air intake with a piece of material will effectively extinguish an intake fire if a chemical extinguisher is not available.

NOTE 2: Where an extinguisher has been used on or around the engine, on no account is a re-start to be attempted until the engine has been examined and certified as serviceable.

RE-STARTING THE ENGINE IN FLIGHT

WITH PROPELLER STOPPED

(1) Assume and trim for the glide.

Check:

Fuel	- ON
Fuel Pump	- ON
Mixture	- RICH
Pitch	- FULLY FINE
Ignition Master Switch	- BOTH ON
Throttle	- SET 1/2 in. OPEN

(2) Engage starter and release immediately engine fires. Re-introduce [illegible] throttle as necessary.

NOTE: As carburetter icing may have been a contributing factor to engine stopping, use carburetter heat as necessary to avoid any further icing.

SECTION 4 - NORMAL PROCEDURES

PRE-START CHECK

(1) After entering the cockpit - Strap in.

(2) Check from left to right around the cockpit:

(a)	Brakes	- Serviceable and parked.
(b)	Throttle	- Freedom of movement - CLOSED. Adjust friction nut.
(c)	Mixture	- FULLY RICH.
(d)	Pitch	- FULLY FINE.
(e)	Fire Extinguishers	- Serviceable and Secure.
(f)	Ignition Switches	- OFF.
(g)	Radio	- OFF.
(h)	Carburetter Air Intake Control	- COLD.
(i)	Flaps	- UP.
(j)	Fuel Cock	- OPEN.
(k)	Fuel Contents Gauge	- Serviceability and accuracy.
(l)	Instruments	- Check.
(m)	Gyros	- Cage.

STARTING - COLD

Preparation

Position the aircraft into wind, with chocks fitted and parking brakes ON.
Aircraft refuelled (water drained after refuelling) and oil tank serviced as required.
External fire extinguisher available.

Starting Procedure

(1)	Fuel	- ON
(2)	Master Switch	- ON
(3)	Auxiliary Pump	- ON
(4)	Generator Switch	- ON
(5)	Prime Engine	- Push the throttle (or priming knob, [illegible] 1-3 strokes depending on engine temperature.
(6)	Throttle	- Slightly open (approximately 1/2 in. [illegible]
(7)	Mixture	- RICH
(8)	Carburetter Heat	- COLD
(9)	Ignition	- [illegible]
(10)	Starter Button	- PUSH [illegible], allow at least [illegible] before selecting
(11)	Ignition	- LEFT MAG. ('BOTH' when firing regularly)

WARMING UP - after STARTING

As soon as the engine starts, adjust throttle to obtain 800 RPM and observe the following:

(1) Watch for oil pressure rise within 30 secs.
Normal oil pressure: 60 - 85 psi.

(2) Do not allow engine speed to exceed 800 RPM for the first minute. Cold oil will not circulate quickly enough for adequate lubrication at higher speeds.

CAUTION: If oil pressure is not indicated within 1/2 minute, stop engine and investigate.

(3) Open throttle to 1200 RPM.
Do not exceed this speed until oil temperature has reached 60°C.

(4) Limit ground running to 4 mins in cold weather and to 2 mins at temperatures above 70°F.

TESTING THE ENGINE - GROUND CHECKS

(Lycoming Service Instruction No. 1113)

(1) Advance throttle to 2000 RPM and check magnetos, drop should not exceed ~~125~~ 175 RPM on either magneto.

(2) Check:

Oil pressure	- 60 - 85 psi
Cylinder temperature	- 260°C (500°F) max.
Oil temperature	- 118°C (245°F) max.

(3) Check operation of carburetter heat control by pulling to HOT and observe decrease in RPM.

(4) Exercise propeller pitch control twice through full range to circulate oil.

START - HOT

Initial procedure is similar to cold start but do not prime engine.
Set controls and push starter button.
If engine does not fire during the first revolution, give one slow pump on throttle, while engine is turning, being careful to return throttle to nearly closed position as soon as engine fires.

(TAKE-OFF CHECK INITIAL ACTIONS)

T	Trim	- set to [illegible] position
	Throttle nut	- Finger TIGHT
M	Mixture	- Fully RICH
	Carb. Air	- COLD
P	Pitch	- Fully FINE
F	Fuel	- ON, check contents
F	Flaps	- As required
I	Ignition	- BOTH
	Instruments	- Check oil pressure and temp.
H	Hatches and Harness	- Canopy closed, harness tight
C	Controls	- Check again for full and free movement

TAKE-OFF AT 1900 LB AUW

NORMAL TAKE-OFF

(1) Select 50% flap (2 notches), release the brakes and then open the throttle smoothly to the fully forward position.
(2) The nosewheel should be lifted off the ground at 40-50 knots (IAS).
(3) The aircraft should be flown off at approximately 60 knots. Allow the airspeed to build up to 70 knots (IAS) before climbing.
(4) At approximately 300 feet flaps may be raised and airspeed adjusted to 80 knots (IAS).
(5) Maintain airspeed at 80 knots (IAS) and continue to climb to full throttle.
(6) Fuel pump OFF.

SHORT - PRECAUTIONARY TAKE-OFF

(1) For a short take-off 50% flap (2 notches) must be used. The shortest take-off run will be obtained if the aircraft is held against the brakes until the application of power is nearly complete.
(2) A speed of 65 knots (IAS) which gives the best angle of climb should be maintained until clear of all obstacles. Speed should then be increased to normal climb speed of 80 knots (IAS) after raising flaps.

CROSSWIND TAKE-OFF

The maximum crosswind component in which the aircraft has been demonstrated to have safe handling characteristics for take-off is 20 knots.

CLIMBING

(1) The speed for maximum rate of climb is 80 knots (IAS) (SL to 5,000 ft) and 75 knots thereafter.
(2) Climbing may be safely accomplished at full throttle.
(3) Rich mixture should always be used for climbing unless leaning is necessary to eliminate rough running due to an over-richness at altitude normally occurring above 5,000 ft.

GENERAL FLYING

FLYING CONTROLS

The controls are well harmonised and the aircraft is easy and pleasant to fly. The controls remain light and effective throughout the speed range, except that the ailerons tend to become heavier as the limiting speed is approached and in inverted flight a high forward stick force is required.

CHANGES OF TRIM

Flaps down - Slightly nose-down.
Flaps Up - Slightly nose-up.
Changes of power and speed promote slight changes in directional trim.

STABILITY

The aircraft is easy to trim under all conditions of flight and holds its trimmed speed well.

FLYING AT REDUCED SPEEDS

Reduce speed to 70 knots (IAS) and apply 50% flap (2 notches), then adjust power to maintain an IAS of approximately 65 knots. Stalling speed under these conditions is about 53 knots (IAS).

CARBURETTER ICING

(1) The effect of ice on this engine is more pronounced than on other engines, it is therefore recommended that the air intake control should be set in the HOT position in outside air temperatures below 40°F.
(2) If the control is not set in the HOT position and COLD air is used, icing may occur, indicated by rough running or loss of power.
(3) If icing does not clear shortly after selection of HOT air, manipulation of the throttle may assist.

MIXTURE CONTROLS

(1) The mixture control should be set to fully rich for starting, ground running, take-off and landing.
(2) Changes in altitude will require corresponding adjustments of the mixture control, normally not below 5,000 ft.
It should always be returned to the fully rich position before commencing a dive and/or aerobatics.
(3) The engine should never be allowed to run for any length of time at reduced RPM where the reduction of RPM is obtained by use of the mixture control.

CRUISING

(1) Power - 2400-2500 RPM
(2) Trim Tab - Adjust
(3) Mixture - Lean

STALLING

(1) The approximate power off stalling speeds (Knots - IAS) are:

At 1900 lb AUW

Flaps Up	55
50% Flaps (2 notches)	52
75% Flaps (3 notches)	51
100% Flaps (4 notches)	48

(2) There is no obvious pre-stall airframe buffet. Stall warning in all cases consists of an aural warning 5-10 knots before the stall occurs.

(3) With power off at the stall, the nose drops gently, with a slight tendency for the nose to oscillate in pitch.

(4) With power on, there is a well defined tendency for a wing to drop.

(5) Stalling speeds are reduced by about 2-3 knots with 1 pilot only in the aircraft.

(6) The stall in a steep turn is normally characterized by a tendency for the aircraft to flick out of the turn and unstall.

(7) Recovery in all cases is straight forward and easy.

(8) Maximum height loss in normal recovery from power off stall is 200 feet.

SPINNING

(1) One and a half turns only are permitted at all weights.

(2) Recommended entry technique:

With the aircraft trimmed, power off, flaps up, gradually raise the nose above the horizon with the wings level.

At the point of stall, apply full rudder in direction of the desired spin, at the same time holding the control column centrally hard back. The aircraft will perform an unstable spin with the associated buffet and speed increase.

Rate of rotation will be relatively high with a steep nose down attitude. Some aircraft may tend to recover on their own accord after 1 full turn.

(3) Recovery: Use standard technique. Recovery is almost immediate.

(4) Inadvertent Spin:

In the event of inadvertent spin, ensure flaps up, throttle closed.

Apply standard recovery technique.

Recovery is almost immediate.

(5) Intentional spins with flaps extended are prohibited.

DIVING

(1) Before commencing a dive, the mixture control should be in the FULLY RICH position.

(2) With the increase in speed the aircraft can be easily held in the dive to the limiting speed without re-trimming.

Directional trim can be maintained by progressive application of rudder.

AESL Airtourer T6
22.9.70

AEROBATICS

Aerobatic manoeuvres may be performed in both categories as set out below. Speeds are recommended entry speeds, knots IAS.

Barrel Roll	115	Loop	130
Slow Roll	90	Stall Turn	130
Aileron Roll	100	Steep Turn over 60°	120, AUW > 1800 lb
Flick Roll	80	Steep Turn over 60°	140, AUW ≤ 1800 lb
Half Roll and Dive Out	85	Chandelle and Lazy Eight	110
Half Loop and Roll Out	140	Inverted Flight	100

PRE-LANDING CHECK

Down-wind leg

B	Brakes	- OFF
U	Undercarriage	- Not applicable
M	Mixture	- RICH
P	Pitch	- FINE
F	Fuel Pump	- ON
F	Flap	- As required
H	Harness	- TIGHT

APPROACH AND LANDING

Use carburettor air heat as required.

GLIDE APPROACH

Maintain a minimum operating speed of 70 knots (IAS) throughout.

POWERED APPROACH

Maintain 70 knots (IAS) throughout, adjust power as required. Prior to reducing speed, the flaps should be fully lowered.

SHORT - PRECAUTIONARY LANDING

Proceed as for powered approach, maintaining airspeed [illegible] speed to not less than 60 knots (IAS) crossing the threshold [illegible]

GOING AROUND AGAIN

(1) At full throttle the aircraft will climb away easily with the flaps fully down.
(2) Open the throttle fully and smoothly.
Carb. air - Cold.
Re-trim as necessary.
Climb at 65 knots (IAS) with the flaps lowered.

(3) At a safe height, about 300 feet, raise the flaps in easy stages.
There is little change in trim and no sink.
Allow the speed to build up to 70 knots (IAS) when flaps are fully up.

CROSSWIND LANDING

MAXIMUM CROSSWIND COMPONENT

The maximum crosswind component in which the aircraft has been demonstrated to have safe handling characteristics for take-off and landing is 20 knots.

CHECK AFTER LANDING

When clear of landing area check:

Trim	- CENTRAL
Flaps	- UP
Fuel Pump	- OFF

STOPPING THE ENGINE

(1) Allow the engine to idle at 800-900 RPM for approximately 2 minutes.
(2) Pull mixture control to IDLE CUT-OFF - do not open throttle, as mis-firing may result.
(3) After engine stops firing, set ignition switch to OFF.

CHECK AFTER STOPPING ENGINE

(1)

Throttle	- CLOSED
Fuel	- OFF
Generator	- OFF
Auxiliary Fuel Pump	- OFF
Radio	- OFF
Gyro Instruments	- CAGED
Master Switch	- OFF

POSITION ERROR CORRECTION TABLE

POWER ON IAS KNOTS

Flap Setting	60-69	70-79	80-89	90-109	110-119	120-129	130-139	140-150
Clean	+3	+3	+3	+2	+1	0	-2	-4
50% 2 notches	+3	+2	+1	-	-	-	-	-
100% 4 notches	+2	+1-1/2	+1	-	-	-	-	-

To obtain CALIBRATED AIRSPEED, ADD above corrections to Indicated Airspeed.

SECTION 5 - PERFORMANCE

The performance information in this section:

(1) Relates to a standard aeroplane fitted with:

(a) ENGINE	: Lycoming 0-320-E1A	
(b) PROPELLER	: Type	: Hartzell HC-C2YL-1B/7663-4.0
	Pitch	: 12°12' - 35° at 30 in station
	Diameter	: 72.0 in.
(c) EXTERNAL MODIFICATIONS	: Nil.	

(2) Is no longer applicable if:
   (a) A propeller of another approved type having different diameter and/or pitch is fitted.
   (b) Other external modifications (excluding radio aerials and navigation lights) are incorporated.

For the purpose of complying with the take-off and landing distance requirements of Civil Aviation Safety Order Nr 4A as currently issued and as subsequently amended or re-issued from time to time this aeroplane is classified in the following Performance Groups.

Group 4 for AUW not exceeding 1900 lb.

The field performance data in the following tables may also be used in establishing compliance with the requirements of CASO 4A.

FIELD PERFORMANCE

MEASURED TAKE-OFF DISTANCE TO 50 FEET

Zero wind, zero slope, hard runway. Unfactored distances in feet at an AUW of 1900 lb.

Ambient Temp. / Altitude	ISA	ISA +5°C	ISA +10°C	ISA +15°C	ISA +20°C
Sea level	1380	1445	1510	1560	1680
1000 ft	1490	1565	1640	1730	1825
2000 ft	1625	1700	1780	1880	1980
3000 ft	1780	1870	1955	2065	2180

WEIGHT CORRECTION

For every 50 lb for which the take-off weight is less than the maximum of 1900 lb, subtract 35 ft from the measured take-off distance, down to a minimum weight of 1500 lb.

WIND CORRECTION

For every knot measured headwind component up to a maximum of 8 knots, subtract 20 ft from the take-off distance.

CORRECTION FOR GRASS SURFACE

For operation from short dry grass add 5% to the take-off distance corrected for wind and weight.

MEASURED LANDING DISTANCE FROM 50 FEET

Zero wind, zero slope, hard runway. Unfactored distances in feet at an AUW of 1900 lb.

Ambient Temp. / Altitude	ISA	ISA +5°C	ISA +10°C	ISA +15°C	ISA +20°C
Sea level	1135	1155	1175	1200	1220
1000 ft	1170	1190	1210	1235	1260
2000 ft	1204	1225	1250	1270	1295
3000 ft	1245	1270	1290	1320	1345

WEIGHT CORRECTION

No reduction in landing distance is permissible at lower weights.

WIND CORRECTION

For every knot measured headwind component up to a maximum of 8 knots subtract 25 ft from the landing distance.

CORRECTION FOR GRASS SURFACE

For operation from short, dry, grass add 10% to the landing distance corrected for wind and weight.

APPENDIX 1 - WEIGHT AND LOADING DATA

The following table summarizes the weights, lever arms, and moments of the various items of loading.

Lever arms are positive for items aft of the datum.

ITEM	WEIGHT (lb)	ARM (in.)	MOMENT (lb. in.)
EMPTY WEIGHT OF AEROPLANE DDQ/71/7/1 (See relevant Equipment List)	1353.8	+79.98	108282.3
PILOT AND PASSENGER, each	170	+90.5	+15,385
FUEL: Total capacity 29.0 Imp. gals (at 7.2 lb/gal)	209	+98.0	+20,482
OIL: Total capacity 1.67 Imp. gals (at 9.0 lb/gal)	15.2	+29.0	+ 441
LUGGAGE: 100 lb, maximum	100	+111.0	+11,100

LOAD LIMITATIONS:

LUGGAGE COMPARTMENT: Maximum permissible load 100 lb. Baggage must not be carried when performing aerobatics.

WEIGHING POSITION: The canopy sills immediately beyind the windscreen are to be used for both lateral and longitudinal levelling.

APPENDIX 1 - WEIGHT AND LOADING DATA

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Lever arms are positive for items aft of the datum.

ITEM	WEIGHT (lb)	ARM (in.)	MOMENT (lb.in.)
EMPTY WEIGHT OF AEROPLANE DDQ/71/7/1 (See relevant Equipment List) DDQ/71/7/2	1340·8 1459·8	+80·81 +81·82	+108351·0 +119442·99
PILOT AND PASSENGER, each	170	+ 90.5	+15,385
FUEL: Total capacity 29.0 Imp. gals (at 7.2 lb/gal)	209	+ 98.0	+20,482
OIL: Total capacity 1.67 Imp. gals (at 9.0 lb/gal)	15.2	+ 29.0	+ 441
LUGGAGE: 100 lb, maximum	100	+111.0	+11,100

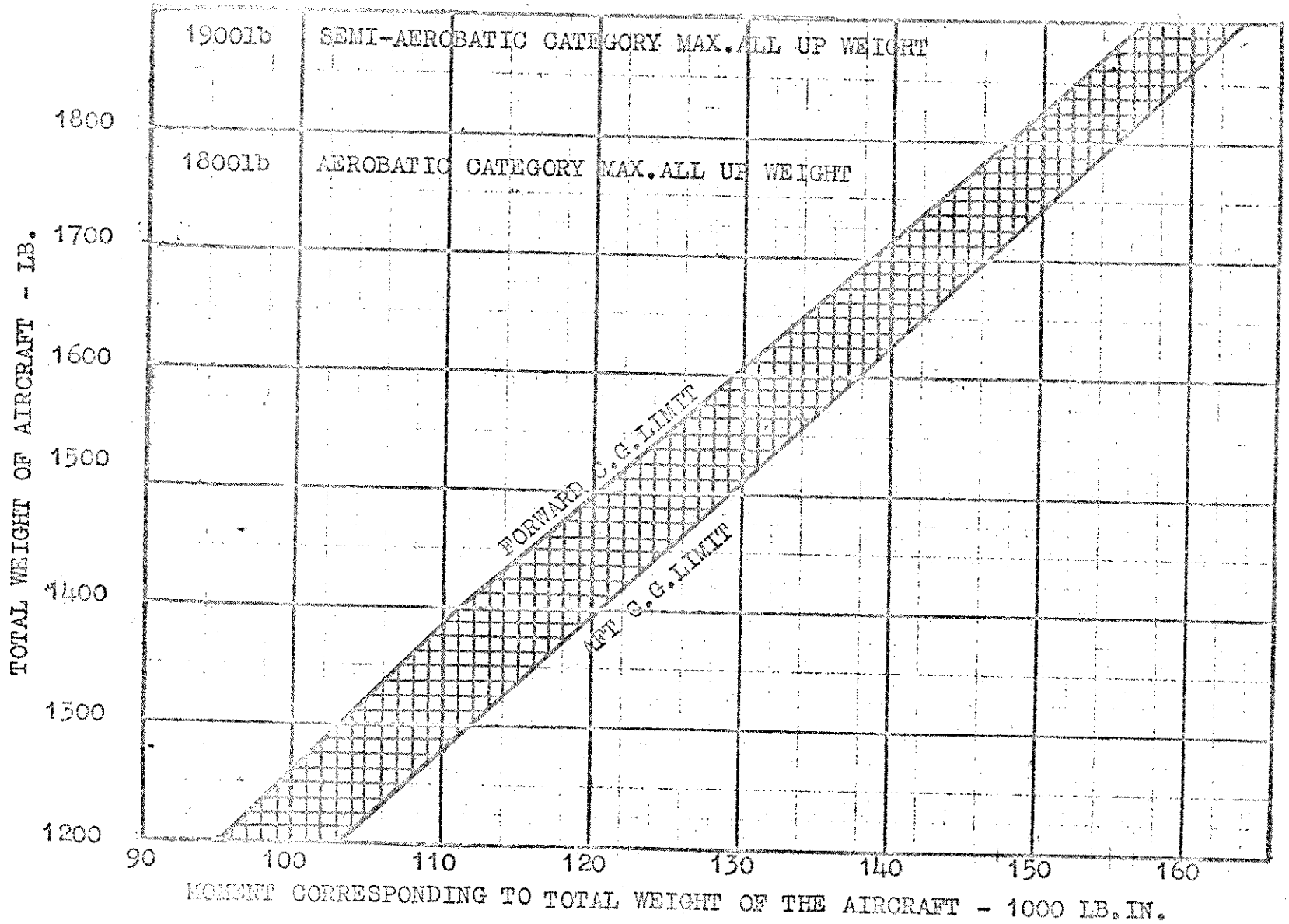
LOAD LIMITATIONS:

LUGGAGE COMPARTMENT: Maximum permissible load 100 lb.
Baggage must not be carried when performing aerobatics.

WEIGHING POSITION: The canopy sills immediately beyind the windscreen are to be used for both lateral and longitudinal levelling.

(1) Add the weight and moment respectively of the empty aeroplane (see loading data) to the weights and moments of all the items to be carried.
(2) Plot this point obtained from (1) on the loading envelope below.
(3) Plot also the points corresponding to all other in-flight weight and moment combinations due to use of fuel or dropping of items in flight.

NOTE Only points falling within the envelope are within the aircraft C.G. limits.



LOADING ENVELOPE

AESL Airtourer T6
22.10.71.
A/L1.

EQUIPMENT LIST DDQ/71/7/1

AIRCRAFT TYPE AIRTOURER T6/24

AIRCRAFT REGISTRATION: 9V-BDQ Serial No. B570

The Aircraft Empty Weight is 1353.8 lbs. with a Corresponding C. of G. Position of +79.98 ins. aft of the Datum.

The items marked O, in the Table below, are included in the Aircraft Empty Weight figure.

Items marked X were not installed in the aircraft when it was weighed and due allowance should therefore be made if they are subsequently installed.

O	First-Aid Kit	O	Clock
O	Fire Extinguisher	O	Carburettor Air Temperature Gauge
O	Axe	X	Undercarriage Position Indicator
O	Airspeed Indicator	O	Flap Position Indicator
O	Altimeter	O	Battery
O	Compass	O	Generator
O	Fuel Contents Gauge	O	Starter
O	Engine Speed Indicator	O	Ammeter
O	Oil Pressure Gauge	O	Radio Type:
O	Fuel Pressure Gauge	O	Vacuum Gauge
O	Manifold Pressure Gauge	O	Intercomm panel
O	Oil Temperature Gauge	O	King KX-175 VHF
O	Cylinder Head Temperature Gauge	O	King K1-203
O	Rate of Climb & Descent Indicator	O	Flap Actuator
O	Turn and Bank Indicator	O	Elevator Trim Actuator
O	Gyro Direction Indicator	O	Flap Indicator
O	Gyro Horizon Indicator	O	Trim Indicator
O	Outside Air Temperature Gauge	O	ADF Equipment

Aircraft last weighed: 6th JULY 1971

Signature:

Date : 10-9-82

FERRY CONFIGURATION

EQUIPMENT LIST DDC/71/7/2

AIRCRAFT TYPE AGSL / EUROPRIP 96/24

AIRCRAFT REGISTRATION: ZK-DDQ Serial No. B570

The Aircraft Empty Weight is 1481.3 lbs. with a Corresponding C. of G. Position of 81.92 ins. aft of the Datum.

The items marked O, in the Table below, are included in the Aircraft Empty Weight figure.
Items marked X were not installed in the aircraft when it was weighed and due allowance should therefore be made if they are subsequently installed.

O	First-Aid Kit	O	Vacuum gauge
O	Fire Extinguisher	O	Intercom panel
O	Axe	O	Mod AES/AT/1027 T.O.F.F. pack
O	Airspeed Indicator		
O	Altimeter		
O	Compass		
O	Fuel Contents Gauge		
O	Engine Speed Indicator		
O	Oil Pressure Gauge		
O	Fuel Pressure Gauge		
O	Manifold Pressure Gauge		
O	Oil Temperature Gauge		
O	Cylinder Head Temperature Gauge		
O	Rate of Climb & Descent Indicator		
O	Turn and Bank Indicator		
O	Gyro Direction Indicator		
O	Gyro Horizon Indicator		
O	Outside Air Temperature Gauge		
O	Clock		
O	Carburettor Air Temperature Gauge		
X	Undercarriage Position Indicator		
O	Flap Position Indicator		
O	Battery		
O	Generator		
O	Starter		
O	Ammeter		
O	Radio Type:		

Aircraft last weighed: 6th JULY 1971

Signature: [illegible]

Date : 6-7-71

FERRY CONFIGURATION

EQUIPMENT LIST DOC/21/7/2

AIRCRAFT TYPE AVRO LINCOLN B2/34

AIRCRAFT REGISTRATION: ZK-DOQ Serial No. B570

The Aircraft Empty Weight is 14818 lbs. with a Corresponding C. of G. Position of 81.82 ins. aft of the Datum.

The items marked O, in the Table below, are included in the Aircraft Empty Weight figure.
Items marked X were not installed in the aircraft when it was weighed and due allowance should therefore be made if they are subsequently installed.

O	First-Aid Kit	O	Vacuum pump
O	Fire Extinguisher	O	Intercomm panel
O	Axe	O	Mod AES/AT/1027B T.O.F.F. pack
O	Airspeed Indicator		
O	Altimeter		
O	Compass		
O	Fuel Contents Gauge		
O	Engine Speed Indicator		
O	Oil Pressure Gauge		
O	Fuel Pressure Gauge		
O	Manifold Pressure Gauge		
O	Oil Temperature Gauge		
O	Cylinder Head Temperature Gauge		
O	Rate of Climb & Descent Indicator		
O	Turn and Bank Indicator		
O	Gyro Direction Indicator		
O	Gyro Horizon Indicator		
O	Outside Air Temperature Gauge		
O	Clock		
O	Carburettor Air Temperature Gauge		
X	Undercarriage Position Indicator		
O	Flap Position Indicator		
O	Battery		
O	Generator		
O	Starter		
O	Ammeter		
O	Radio Type:		

Aircraft last weighed: 6th JULY 1971

Signature: [illegible]

Date : 6-7-71

APPENDIX 2 - MODIFICATIONS

When any of the approved modifications listed below are embodied in this aircraft the basic information in this Flight Manual is subject to amendment as detailed.

INDEX OF MODIFICATIONS

AES/AT/1001 INSTALLATION OF BENDIX CNS 220 VHF SYSTEM
(Aero Engine Services Ltd)

SEEL ENGINEERING INSTRUCTION NO. 93 AND SEEL DRAWING NO. AV/1169 - **REPLACEMENT OF NARCO COM/NAV MARK 12B WITH KING KX-175B AND REWORK OF AUDIO DISTRIBUTION.**

(Amendment No. A/L 4)

MODIFICATION AES/AT/1001

INSTALLATION OF BENDIX CNS 220 VHF SYSTEM TYPE RT-221A-14 RECEIVER/TRANSMITTER

(AERO ENGINE SERVICES LTD)

GENERAL

This Modification provides for the installation of a Bendix CNS 220 VHF system, type RT-221A-14 receiver transmitter.

WEIGHT AND LOADING DATA

The following weight and loading data which covers the installation of AES/AT/1001 must be taken into account when determining the C of G of the loaded aircraft.

However, if the equipment list is marked to indicate that the modification was installed when the aircraft was last weighed the figure relevant to the "Resultant Change" due to the incorporation of the modification must be disregarded.

	WEIGHT (lb)	ARM (in)	MOMENT (lb.in.)
Installation of Bendix receiver/transmitter	7	+63	+441

MODIFICATION

SEEL ENGINEERING INSTRUCTION NO. 93

AND SEEL DRAWING NO. AV/1169

(SEEL AVIATION DIVISION)

GENERAL

THIS MODIFICATION CALLS FOR REMOVAL OF EXISTING NARCO MARK 12B TRANSCEIVER WITH T(24) MP-12A-1 POWER UNIT AND VOA-8 CONVERTER - INDICATOR AND INSTALL KING KX-175 TRANSCEIVER WITH KI-203 CONVERTER - INDICATOR AND KA-39 VOLTAGE CONVERTER. REWORK AND WIRE THE AUDIO PANEL.

WEIGHT AND LOADING DATA

THE FOLLOWING WEIGHT AND LOADING DATA WHICH COVERS IN THE ABOVE MODIFICATION MUST BE TAKEN INTO ACCOUNT WHEN DETERMINING THE C OF G OF THE LOADED AIRCRAFT.

HOWEVER, IF THE EQUIPMENT LIST IS MARKED TO INDICATE THAT THE MODIFICATION WAS INSTALLED WHEN THE AIRCRAFT WAS LAST WEIGHED THE FIGURE RELEVANT TO THE "RESULTANT CHANGE" DUE TO THE INCORPORATION OF THE MODIFICATION MUST BE DISREGARD.

	WEIGHT(lbs)	ARM	MOMENT
REMOVED NARCO MARK 12B	5.7	66	376.2
T(24) MP-12A-1	3.9	154	600.6
VOA-8	2.85	66	188.1
	12.45		1164.9
INSTALLED KING KX-175B	7.4	66	488.4
KI-203	1.6	66	105.6
KA-39	1.2	51	61.2
	10.2		655.2

MODIFICATION

SEEL ENGINEERING INSTRUCTION NO. 93

AND SEEL DRAWING NO. AV/1169

(SEEL AVIATION DIVISION)

GENERAL

THIS MODIFICATION CALLS FOR REMOVAL OF EXISTING NARCO MARK 12B TRANSCEIVER WITH T(24) MP-12A-1 POWER UNIT AND VOA-8 CONVERTER - INDICATOR AND INSTALL KING KX-175 TRANSCEIVER WITH KI-203 CONVERTER - INDICATOR AND KA-39 VOLTAGE REGULATOR. REWORK AND WIRE THE AUDIO PANEL.

WEIGHT AND LOADING DATA

THE FOLLOWING WEIGHT AND LOADING DATA WHICH COVERS IN THE ABOVE MODIFICATION MUST BE TAKEN INTO ACCOUNT WHEN DETERMINING THE C OF G OF THE LOADED AIRCRAFT.

HOWEVER, IF THE EQUIPMENT LIST IS MARKED TO INDICATE THAT THE MODIFICATION WAS INSTALLED WHEN THE AIRCRAFT WAS LAST WEIGHED THE FIGURE RELEVANT TO THE "RESULTANT CHANGE" DUE TO THE INCORPORATION OF THE MODIFICATION MUST BE DISREGARD.

	WEIGHT(lbs)	ARM	MOMENT
REMOVED NARCO MARK 12B	5.7	66	376.2
T(24) MP-12A-1	3.9	154	600.6
VOA-8	2.85	66	188.1
	12.45		1164.9
INSTALLED KING KX-175B	7.4	66	488.4
KI-203	1.6	66	105.6
KA-39	1.2	51	61.2
	10.2		655.2

CHART C—BASIC WEIGHT AND BALANCE RECORD

(CONTINUOUS HISTORY OF CHANGES IN STRUCTURE OR EQUIPMENT AFFECTING WEIGHT AND BALANCE)

FOR USE IN T. O. 1-1B-40 & AN 01-1B-40

AIRPLANE MODEL: AESL Airtourer

SERIAL NO.: 9V-BDQ

PAGE NO.: 1

DATE	ITEM NO. IN	ITEM NO. OUT	DESCRIPTION OF ARTICLE OR MODIFICATION	WEIGHT CHANGE ADDED (+) WEIGHT Lb	ADDED (+) ARM In	ADDED (+) MOMENT[1] Lb In	REMOVED (−) WEIGHT Lb	REMOVED (−) ARM In	REMOVED (−) MOMENT[1] Lb In	RUNNING TOTAL BASIC AIRPLANE WEIGHT	MOMENT[1]	C.G. In
6-7-71			Weight and Balance as weighed and computed by SGAS (Report No. DDQ/71/7/1)	-	-	-	-	-	-	1356	108792	80.23
30-10-81		X	VOR/LOC Converter/Indicator Type VOA-8 S/N 23BK1	-	-	-	2.85	66	188.1			
30-10-81		X	NAV/COMM Unit Type NARCO Mk 12B	-	-	-	5.7	66	376.2			
30-10-81		X	Power/Audio Unit Type T-24MP-12A1	-	-	-	3.9	154	600.6			
30-10-81	X		Indicator P/N 066-3034-00	1.6	66	105.6	-	-	-			
30-10-81	X		NAV/COMM SYS P/N 069-1019-00	7.4	66	488.4	-	-	-			
30-10-81	X		Power Converter P/N 071-1041-01	1.2	51	61.2	-	-	-			
			Weight & Balance as computed by SAMCO - Albert Wong Fatt Sang 30/10/81	10.2		655.2	12.45		1164.9	1353.8	108282.3	79.98

24.11.81

[1] Enter constant used below line.

[2] Balance computer index.

DD FORM 1 SEP 54 365 C

PREVIOUS EDITIONS OF THIS FORM MAY BE USED UNTIL STOCKS ARE EXHAUSTED.

S/N 0102-003-8300

SINGAPORE GENERAL AVIATION SERVICE COMPANY PTE. LTD.

WEIGHT AND CENTRE OF GRAVITY SCHEDULE

Ref: DOQ/71/7/1 Produced By: A.E.S.L. Aircraft Designation: AIRTOURER T6/24

Nat. & Regr Marks: 9V-BDQ Ser.No: B571 Max Permissible Weight: 1900 LB

C of G limits for Aerobatic Operation: MAX WEIGHT 1800 LBS

The load must be distributed so that the C of G lies between a forward limit of 79.2 ins. aft of datum at 1400 lbs. or less and 81.5 ins. aft of datum at 1750 lbs. with linear interpolation at intermediate weights and an aft limit of 86.1 ins. aft of datum at all weights. TO CALCULATE C OF G LIMIT AT 1800 LBS INTERPOLATE BETWEEN 81.5 AT 1750 AND 84.0 AT 1900

C of G limits for Normal Operation:

The load must be distributed so that the C of G lies between a forward limit of 79.2 ins. aft of datum at 1400 lbs. or less and 82.5 ins. aft of datum at 1900 lbs. with linear interpolation at intermediate weights and an aft limit of 86.1 ins. aft of datum at all weights. USING 81.5 AT 1750 LBS AS AN INTERMEDIATE POINT

Part A. Basic Weight:

The basic weight of the aircraft as calculated from weighing Report: DOQ/71/7/1 Dated: 6.7.71 is 1356 lbs.

The C of G of the Aircraft in the same condition at this weight, (landing gear extended) is (aft of datum) 80.23 ins.

The total moment about the datum in this condition in lb. ins/100 is 1087.92

The datum referred to is defined in the C of A validation or related flight manual is:

The basic weight includes the weight of total unusable fuel and oil and the weight of the following comprising the list of "Basic Equipment":-

1) FIRST AID KIT
2) FIRE EXTINGUISHER
3) [illegible]

Part B. Variable Load:

The weight and lever arms of the variable load are shown below. The variable load depends upon the equipment carried for the particular role and includes the weight of crew and usable fluids other than fuel and items of equipment not forming part of basic equipment:-

ITEM	WT. LBS.	LEVER ARM INS.	MOMENT CHANGE LBS. INS/100
PILOT	170	90.5	153.85

Part C. Operating Weight (A.P.S.)

The operating or (Aircraft prepared for service) weight and C of G assuming that all the variable load specified in part B is carried (log gear extended) are:-

	Weight lb.	Lever Arm ins	Moment lb.ins/100
AEROBATIC ROLE:- / NORMAL ROLE :-	1526	81.37	1241.77

The operating weight is defined as the sum of the basic wt. and wt. of variable load.

Part D. Loading Information (Disposable load)

Information is given below to enable the disposable load (fuel and Payload) to be distributed so that the maximum weight and C of G limits defined in the C of A/validation or flight manual are not exceeded.

The total moment change when the landing gear is retracted is lb.ins./100

ITEM		WEIGHT	LEVER ARM INS.
Fuel in tanks main.	Tot. usable. 29 IMP.Gal @ 7.1 lbs/Gal	205.9	+98.0
Fuel in tanks aux.	Tot. usable. IMP.Gal @ 7.1 lbs/Gal		
Oil in tanks	Tot. usable. 1.67 IMP.Qts @ 9 Lbs/Gal	15.2	+29.0

PASSENGERS IN SEATS. ROW 1 170 LBS LEVER ARM +90.5
" " ROW 2 .....
" " ROW 3 .....
" " ROW 4 .....
" " ROW 5 .....
" " ROW 6 .....
" " ROW 7 .....
" " ROW 8 .....

Baggage REAR COMPT. 100 LBS MAX LEVER ARM +111.0
Baggage .....
Baggage NOTE. BAGGAGE MUST NOT BE CARRIED WHEN DOING AEROBATICS

Check relevant manufacturers loading instructions

NOTE: The total loaded weight of the aircraft is the sum of the operating weight and the weight of the disposable load.

This schedule was prepared on date 14.9.71 and supercedes all previous issues.

SIGNED: .....

FOR AND ON BEHALF OF: .....